



LIGHT WAVE DEHAZE

Team Members

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Problem Statement

$$I(x) = J(x)t(x) + A\big[1 - t(x)\big]$$



Hazy image I



Clean image J

- Recover a clean image from a single hazy observation.

Why Dehazing Is Needed

Haze obscures distant objects



Preprocessing for detection and tracking

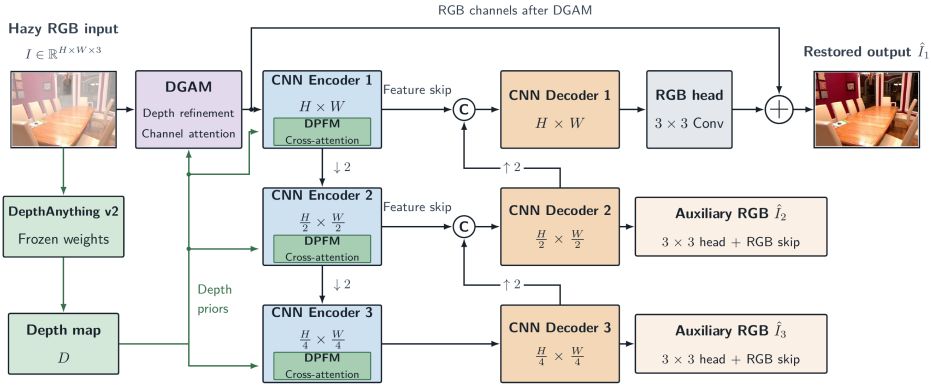


Existing Dehazing Approaches

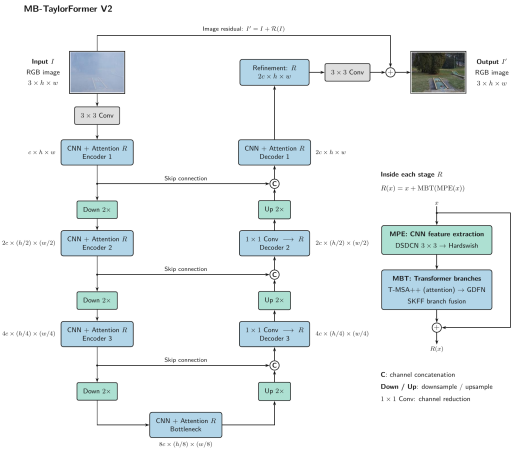
Approach	Core idea	Key consideration
Physical Prior	Restore images using assumptions about haze and the scene	Depend on how well the assumptions match the scene
CNN	Learn features through convolution	Ability to capture wider context
Transformer	Model relationships across image regions	Computational and memory costs
Diffusion	Restore images through denoising	Latency with multiple inference steps

Trade-offs depend on the architecture and its optimization.

UDPNet


RGB channels after DGAM
Hazy RGB input
$I \in \mathbb{R}^{H \times W \times 3}$
Restored output $\hat{I}_1$
DGAM
Depth refinement
Channel attention
CNN Encoder 1
$H \times W$
DPFM
Cross-attention
Feature skip
CNN Decoder 1
$H \times W$
RGB head
3×3 Conv
$\downarrow 2$
$\uparrow 2$
CNN Encoder 2
$\frac{H}{2} \times \frac{W}{2}$
DPFM
Cross-attention
Feature skip
CNN Decoder 2
$\frac{H}{2} \times \frac{W}{2}$
Auxiliary RGB $\hat{I}_2$
3×3 head + RGB skip
DepthAnything v2
Frozen weights
Depth
priors
Depth map
D
CNN Encoder 3
$\frac{H}{4} \times \frac{W}{4}$
DPFM
Cross-attention
CNN Decoder 3
$\frac{H}{4} \times \frac{W}{4}$
Auxiliary RGB $\hat{I}_3$
3×3 head + RGB skip
C: concatenation
$\downarrow 2$ / $\uparrow 2$: down-/upsampling
DPFM: depth prior fusion module

MB-TaylorFormer V2

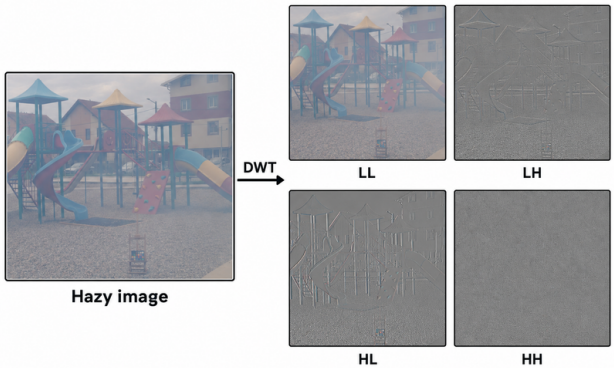


Combining Strengths

How can we use the strengths of each method at different stages of image processing?



Frequency-Domain Observation



Low-frequency component (LL)
Global appearance and scene structure.

High-frequency components
Edges and local details in LH, HL, and HH.

Key idea
Match each method's strengths to its restoration stage.

Overall Design and Objectives



Hazy input

I_{h}

Stage 1

Global restoration

LFRN

Low-frequency recovery

Attention on LM

$\widetilde{I}$

Stage 2

Detail refinement

DEN

Detail enhancement

Spatial + frequency



Restored image

$\widehat{I}$

- **Restoration:** remove global haze while preserving local details.
- **Efficiency:** balance restoration quality with inference cost.

Dataset
SOTS-ITS / RESIDE ITS

- Synthetic indoor hazy images paired with clean reference images.
- **RESIDE ITS:** Indoor Training Set for training and validation.
- **SOTS Indoor:** indoor subset of the Synthetic Objective Testing Set.

12,990	1,000	500
Training images	Validation images	Test images

Train and validate on RESIDE ITS; evaluate restoration quality on SOTS Indoor.

Evaluation Metrics

Metric	PSNR (dB)	SSIM	FPS	QSS
Full name	Peak signal-to-noise ratio	Structural similarity index	Frames per second	Quality-Speed Score
What it measures	Pixel accuracy against a clean reference.	Structure, contrast, and luminance similarity.	Number of images processed per second.	Balance between quality and speed. $\mathrm{QSS} = \frac{\mathrm{FPS}}{\mathrm{MSE}}$
How to interpret it	Higher values indicate lower pixel error.	Values closer to 1 indicate greater similarity.	Higher values indicate faster processing.	Higher values favor faster processing and lower pixel error.

MSE: mean squared error against a clean reference. Use the same image value range for MSE.

Compare FPS and QSS using the same hardware, image resolution, and batch size.

Target

Targets on the SOTS dataset

PSNR	SSIM	QSS
$\geq$ 25 dB	$\geq$ 0.95	$>$ 0.45

- Optimize the frequency-domain and RGB restoration blocks.
- Assess latency, parameters, and FLOPs for future edge deployment.

Proposed targets, not reported results. For QSS, fix the hardware, image resolution, batch size, and MSE image value range.

Checkpoint Results

Method	PSNR (dB)	SSIM	FPS	QSS	Params (M)	FLOPs (G)
DCP	21.70	0.8915	7.8008	0.01774	0	0
CAP	22.76	0.8833	4.1056	0.01192	0	0
GridDehazeNet	33.21	0.9860	7.6970	0.24788	0.958	37.422
MB-TaylorFormerV2	35.61	0.9873	3.6658	0.20516	7.296	186.799
UDPNet	37.54	0.9919	3.0700	0.26795	13.590 + 97.471	246.960 + 380.701
LightWaveDehaze (ours)*	**31.084**	**0.9691**	**17.263**	**0.34075**	**10.137**	**49.286**

$$\mathrm{QSS} = \frac{\mathrm{FPS} \times 10^{\mathrm{PSNR}/10}}{\mathrm{MAX}^2}, \qquad \mathrm{MAX} = 255$$

* Ours: illustrative checkpoint values, not measured results; SSIM retained from the supplied table.
QSS is calculated from the displayed PSNR and FPS using the 8-bit image range.

Next Steps

1. Experiment with Mamba

Replace Attention with Mamba in the low-frequency recovery block (LFRN) and compare it with the current Attention-based baseline.

2. Optimize the corresponding hyperparameters

Tune Mamba state size, expansion factor, and block depth, together with training settings such as learning rate, batch size, and weight decay.

3. Evaluate the quality–speed balance

Compare PSNR, SSIM, FPS, and QSS on the same dataset split and hardware; select the configuration that best meets the target metrics.

Thank You

Questions and Discussion